

A Frobenius-persistent nonvanishing for an ample bundle on $\mathrm{Gr}(2, 4)$

Statement

Let k be a perfect field of characteristic $p > 0$, and let $F = F_X$ denote the absolute Frobenius of a k -scheme X . The purpose of this note is to give a self-contained counterexample to the following plausible vanishing assertion:

$$H^i(X, F^{n*}\mathcal{E} \otimes \mathcal{G}) = 0 \quad \text{for all } i \geq \mathrm{rk}(\mathcal{E}) \text{ and all } n \gg 0,$$

where X is smooth projective, $\mathcal{E}$ is ample, and $\mathcal{G}$ is coherent.

Theorem 1. *Let k be a perfect field of characteristic $p > 0$. There exist a smooth projective Grassmannian $X = \mathrm{Gr}(2, 4)$, an ample vector bundle $\mathcal{E}$ of rank 2 on X , and a coherent sheaf $\mathcal{G}$ on X such that*

$$H^2(X, F^{n*}\mathcal{E} \otimes \mathcal{G}) \neq 0 \quad \text{for every } n \geq 0.$$

In particular, Frobenius pullbacks of ample vector bundles need not eventually kill cohomology in degrees $i \geq \mathrm{rk}(\mathcal{E})$.

The example will be

$$X = \mathrm{Gr}(2, k^4), \quad \mathcal{E} = F^*\mathcal{Q} \otimes \mathcal{O}_X(1), \quad \mathcal{G} = \omega_X,$$

where

$$0 \longrightarrow \mathcal{S} \longrightarrow k^4 \otimes \mathcal{O}_X \longrightarrow \mathcal{Q} \longrightarrow 0$$

is the tautological sequence and $\mathcal{O}_X(1) = \det \mathcal{Q}$.

1 Persistence of nonvanishing on Frobenius-split varieties

We first isolate the mechanism that makes a single nonvanishing persist under all Frobenius pullbacks.

Lemma 2 (Persistence lemma). *Let Y be a smooth projective Frobenius-split variety over k . Let $\mathcal{A}$ be a vector bundle on Y . If*

$$H^i(Y, \mathcal{A} \otimes \omega_Y) \neq 0,$$

then

$$H^i(Y, F^{n*}\mathcal{A} \otimes \omega_Y) \neq 0 \quad \text{for every } n \geq 0.$$

More precisely, after restricting scalars along the n -fold Frobenius of k ,

$$H^i(Y, \mathcal{A} \otimes \omega_Y)$$

is a direct summand of

$$H^i(Y, F^{n*} \mathcal{A} \otimes \omega_Y).$$

Proof. Since Y is Frobenius split, the natural inclusion

$$\mathcal{O}_Y \longrightarrow F_* \mathcal{O}_Y$$

splits as a map of $\mathcal{O}_Y$ -modules. Iterating the splitting gives, for every $n \geq 0$, a decomposition

$$F_*^n \mathcal{O}_Y \simeq \mathcal{O}_Y \oplus \mathcal{R}_n$$

for some $\mathcal{O}_Y$ -module $\mathcal{R}_n$.

Because Y is smooth, Grothendieck duality for the finite flat map $F^n : Y \rightarrow Y$ gives

$$\mathcal{H}om_{\mathcal{O}_Y}(F_*^n \mathcal{O}_Y, \omega_Y) \simeq F_*^n \omega_Y.$$

Equivalently, this is the n -fold Cartier trace pairing. Etale-locally, if $t_1, \dots, t_d$ are parameters on Y , then the local basis elements

$$F_*^n t^\alpha, \quad 0 \leq \alpha_j < p^n,$$

of $F_*^n \mathcal{O}_Y$ pair perfectly with the local basis elements

$$F_*^n (t^{(p^n-1, \dots, p^n-1)-\alpha} dt_1 \wedge \dots \wedge dt_d)$$

of $F_*^n \omega_Y$.

Applying $\mathcal{H}om_{\mathcal{O}_Y}(-, \omega_Y)$ to the direct-sum decomposition of $F_*^n \mathcal{O}_Y$, we see that ω_Y is a direct summand of $F_*^n \omega_Y$. Tensoring with $\mathcal{A}$ and using the projection formula gives $\mathcal{A} \otimes \omega_Y$ as a direct summand of

$$\mathcal{A} \otimes F_*^n \omega_Y \simeq F_*^n (F^{n*} \mathcal{A} \otimes \omega_Y).$$

Taking cohomology proves the direct-summand assertion. The Frobenius twist of the k -vector-space structure is irrelevant for vanishing, so the stated nonvanishing follows. $\square$

2 The Grassmannian and its Frobenius splitting

Let $V = k^4$ and choose a volume form $\det V \simeq k$. Let

$$X = \mathrm{Gr}(2, V)$$

be the Grassmannian of two-dimensional subspaces of V . Let

$$0 \longrightarrow \mathcal{S} \longrightarrow V \otimes \mathcal{O}_X \longrightarrow \mathcal{Q} \longrightarrow 0$$

be the tautological sequence, where $\mathcal{S}$ and $\mathcal{Q}$ both have rank 2, and put

$$\mathcal{O}_X(1) = \det \mathcal{Q}.$$

For any sheaf $\mathcal{H}$ on X , write $\mathcal{H}(t) = \mathcal{H} \otimes \mathcal{O}_X(t)$.

Lemma 3. *The canonical bundle of X is*

$$\omega_X \simeq \mathcal{O}_X(-4).$$

Proof. The tangent bundle is

$$\mathcal{T}_X \simeq \mathcal{S}^\vee \otimes \mathcal{Q}.$$

Since $\det \mathcal{S} \otimes \det \mathcal{Q} \simeq \det V \otimes \mathcal{O}_X \simeq \mathcal{O}_X$, the chosen volume form gives

$$\det \mathcal{S}^\vee \simeq \det \mathcal{Q} = \mathcal{O}_X(1).$$

Therefore

$$\det \mathcal{T}_X \simeq (\det \mathcal{S}^\vee)^{\otimes 2} \otimes (\det \mathcal{Q})^{\otimes 2} \simeq \mathcal{O}_X(4).$$

Thus $\omega_X = \det \Omega_X^1 \simeq (\det \mathcal{T}_X)^{-1} \simeq \mathcal{O}_X(-4)$. □

Lemma 4. *The Grassmannian $X = \mathrm{Gr}(2, V)$ is Frobenius split.*

Proof. We give an explicit splitting. Under the Plucker embedding

$$X \hookrightarrow \mathbb{P}(\wedge^2 V) \simeq \mathbb{P}^5,$$

choose homogeneous coordinates

$$x_0 = x_{12}, \quad x_1 = x_{13}, \quad x_2 = x_{14}, \quad x_3 = x_{23}, \quad x_4 = x_{24}, \quad x_5 = x_{34}.$$

Then X is the smooth quadric hypersurface cut out by

$$f = x_0x_5 - x_1x_4 + x_2x_3.$$

The formula remains valid in characteristic 2, with the evident interpretation of signs.

Let $R = k[x_0, \dots, x_5]$. Let

$$T : F_*R \longrightarrow R$$

be the standard Frobenius trace, characterized on monomials by

$$T(F_*x^\alpha) = \begin{cases} x^\beta, & \text{if } \alpha_j = p\beta_j + (p-1) \text{ for every } j, \\ 0, & \text{otherwise.} \end{cases}$$

This is R -linear for the R -module structure on F_*R .

A homogeneous polynomial P of degree $6(p-1)$ defines an $\mathcal{O}_{\mathbb{P}^5}$ -linear map

$$\varphi_P : F_*\mathcal{O}_{\mathbb{P}^5} \longrightarrow \mathcal{O}_{\mathbb{P}^5}$$

by sheafifying the p^{-1} -linear map

$$F_*a \longmapsto T(F_*(Pa)).$$

Take

$$P = f^{p-1}(x_1x_2x_3x_4)^{p-1}.$$

The coefficient of $(x_0x_1x_2x_3x_4x_5)^{p-1}$ in P is 1: indeed, the only contributing term is obtained by taking $(x_0x_5)^{p-1}$ from f^{p-1} . Hence

$$\varphi_P(F_*1) = 1,$$

so φ_P is a Frobenius splitting of $\mathbb{P}^5$.

It remains to check that this splitting is compatible with the quadric X . Write

$$h = (x_1x_2x_3x_4)^{p-1}, \quad \text{so that} \quad P = f^{p-1}h.$$

For every homogeneous $a \in R$,

$$\varphi_P(F_*(fa)) = T(F_*(f^pha)) = f T(F_*(ha)) \in (f),$$

where the middle equality uses R -linearity of T and the fact that multiplication by f on F_*R is multiplication by f^p before applying F_* . Thus the ideal sheaf of X is preserved by φ_P , and the splitting descends to X . $\square$

3 The ample bundle

Define

$$\mathcal{E} := F^*\mathcal{Q} \otimes \mathcal{O}_X(1).$$

Then $\mathcal{E}$ has rank 2.

Lemma 5. *The vector bundle $\mathcal{E}$ is ample.*

Proof. Pulling back the tautological quotient

$$V \otimes \mathcal{O}_X \twoheadrightarrow \mathcal{Q}$$

by Frobenius gives a quotient from a trivial rank-four bundle onto $F^*\mathcal{Q}$. After tensoring with $\mathcal{O}_X(1)$, we obtain a quotient

$$\mathcal{O}_X(1)^{\oplus 4} \twoheadrightarrow \mathcal{E}.$$

Since $\mathcal{O}_X(1)$ is ample, the vector bundle $\mathcal{O}_X(1)^{\oplus 4}$ is ample, and every quotient of an ample vector bundle is ample. Hence $\mathcal{E}$ is ample. $\square$

The basic nonvanishing we need is

$$H^2(X, \mathcal{E} \otimes \omega_X) \neq 0.$$

By Lemma 3, this is equivalent to

$$H^2(X, F^*\mathcal{Q}(-3)) \neq 0.$$

We now prove this nonvanishing.

4 A rank-two Frobenius exact sequence

Lemma 6. *Let $\mathcal{M}$ be a rank-two vector bundle on a scheme of characteristic $p > 0$. There is a functorial short exact sequence*

$$0 \longrightarrow F^* \mathcal{M} \longrightarrow \mathrm{Sym}^p \mathcal{M} \xrightarrow{\delta} \mathrm{Sym}^{p-2} \mathcal{M} \otimes \det \mathcal{M} \longrightarrow 0.$$

Here $F^* \mathcal{M} \rightarrow \mathrm{Sym}^p \mathcal{M}$ is the linearization of the p -th power map $m \mapsto m^p$.

Proof. Let

$$d : \mathrm{Sym}^p \mathcal{M} \longrightarrow \mathrm{Sym}^{p-1} \mathcal{M} \otimes \mathcal{M}$$

be the comultiplication, or first-derivative, map. Locally, in a basis x, y , it is given by

$$d(x^a y^{p-a}) = ax^{a-1} y^{p-a} \otimes x + (p-a)x^a y^{p-a-1} \otimes y.$$

The multiplication map

$$\mathrm{Sym}^{p-1} \mathcal{M} \otimes \mathcal{M} \longrightarrow \mathrm{Sym}^p \mathcal{M}$$

composed with d is multiplication by p , hence is zero in characteristic p . Thus d factors through the Koszul kernel

$$\ker(\mathrm{Sym}^{p-1} \mathcal{M} \otimes \mathcal{M} \rightarrow \mathrm{Sym}^p \mathcal{M}) \simeq \mathrm{Sym}^{p-2} \mathcal{M} \otimes \det \mathcal{M}.$$

Let

$$\delta : \mathrm{Sym}^p \mathcal{M} \longrightarrow \mathrm{Sym}^{p-2} \mathcal{M} \otimes \det \mathcal{M}$$

be the resulting map.

In a local basis x, y , the inclusion of this Koszul kernel is

$$s \otimes (x \wedge y) \longmapsto sy \otimes x - sx \otimes y.$$

Therefore δ is given locally by

$$\delta(x^a y^{p-a}) = \begin{cases} ax^{a-1} y^{p-a-1} \otimes (x \wedge y), & 1 \leq a \leq p-1, \\ 0, & a = 0 \text{ or } a = p. \end{cases}$$

The coefficients a are units for $1 \leq a \leq p-1$, so δ is locally surjective and its kernel is locally generated by x^p and y^p . This kernel is precisely the image of the canonical map

$$F^* \mathcal{M} \longrightarrow \mathrm{Sym}^p \mathcal{M}, \quad F^* m \longmapsto m^p.$$

Hence the sequence is exact. □

Applying Lemma 6 to $\mathcal{M} = \mathcal{Q}$ and using $\det \mathcal{Q} = \mathcal{O}_X(1)$ gives

$$0 \longrightarrow F^* \mathcal{Q} \longrightarrow \mathrm{Sym}^p \mathcal{Q} \longrightarrow \mathrm{Sym}^{p-2} \mathcal{Q}(1) \longrightarrow 0.$$

After twisting by $\mathcal{O}_X(-3)$, we obtain

$$0 \longrightarrow F^* \mathcal{Q}(-3) \longrightarrow \mathrm{Sym}^p \mathcal{Q}(-3) \longrightarrow \mathrm{Sym}^{p-2} \mathcal{Q}(-2) \longrightarrow 0. \quad (1)$$

5 The required cohomology calculation

We compute the cohomology of the two outer terms in (1) by passing to the flag variety.

Let

$$Y = \text{Fl}(2, 3; V)$$

be the variety of flags $U \subset H \subset V$ with $\dim U = 2$ and $\dim H = 3$. Let

$$\pi : Y \longrightarrow X = \text{Gr}(2, V), \quad g : Y \longrightarrow Z := \text{Gr}(3, V)$$

be the two projections. We use the quotient convention for projective bundles.

Then

$$Y \simeq \mathbb{P}_X(\mathcal{Q}),$$

where the relative $\mathcal{O}(1)$ is

$$\mathcal{O}_\pi(1) = V/H.$$

Similarly,

$$Y \simeq \mathbb{P}_Z(\mathcal{T}),$$

where $\mathcal{T} \subset V \otimes \mathcal{O}_Z$ is the tautological rank-three subbundle on Z and

$$\mathcal{O}_g(1) = H/U.$$

Let

$$L := \mathcal{O}_Z(1) = V/H$$

be the quotient-line Plucker bundle on Z . Then on Y one has

$$\mathcal{O}_\pi(1) = g^*L$$

and

$$\pi^*\mathcal{O}_X(1) = g^*L \otimes \mathcal{O}_g(1),$$

since V/U has determinant $(V/H) \otimes (H/U)$.

Lemma 7. *For every $m \geq 0$,*

$$H^\bullet(X, \text{Sym}^m \mathcal{Q}(-2)) = 0.$$

Moreover, for every $m \geq 0$ and every integer j ,

$$H^j(X, \text{Sym}^m \mathcal{Q}(-3)) \simeq H^{j-2}(Z, L^{m-2}),$$

with the convention that $H^r = 0$ for $r < 0$.

Proof. Because $Y = \mathbb{P}_X(\mathcal{Q})$ with quotient convention,

$$\pi_*\mathcal{O}_\pi(m) = \text{Sym}^m \mathcal{Q}, \quad R^a\pi_*\mathcal{O}_\pi(m) = 0 \quad (a > 0, m \geq 0).$$

Thus, for every integer t ,

$$\begin{aligned} H^j(X, \text{Sym}^m \mathcal{Q}(t)) &\simeq H^j(Y, \mathcal{O}_\pi(m) \otimes \pi^*\mathcal{O}_X(t)) \\ &\simeq H^j(Y, g^*L^{m+t} \otimes \mathcal{O}_g(t)). \end{aligned}$$

For $t = -2$, the fibers of g are projective planes, and

$$H^a(\mathbb{P}^2, \mathcal{O}_{\mathbb{P}^2}(-2)) = 0 \quad \text{for every } a.$$

Therefore

$$R^a g_* \mathcal{O}_g(-2) = 0 \quad \text{for every } a,$$

and it follows that

$$H^\bullet(X, \text{Sym}^m \mathcal{Q}(-2)) = 0.$$

For $t = -3$, relative duality for

$$g : Y = \mathbb{P}_Z(\mathcal{T}) \longrightarrow Z$$

gives

$$R^a g_* \mathcal{O}_g(-3) = 0 \quad (a \neq 2), \quad R^2 g_* \mathcal{O}_g(-3) \simeq (\det \mathcal{T})^{-1}.$$

Since $\det \mathcal{T} \otimes L \simeq \det V \otimes \mathcal{O}_Z \simeq \mathcal{O}_Z$, the chosen volume form identifies

$$(\det \mathcal{T})^{-1} \simeq L.$$

Thus

$$R^2 g_*(g^* L^{m-3} \otimes \mathcal{O}_g(-3)) \simeq L^{m-3} \otimes L \simeq L^{m-2},$$

and all other higher direct images are zero. The Leray spectral sequence therefore gives

$$H^j(Y, g^* L^{m-3} \otimes \mathcal{O}_g(-3)) \simeq H^{j-2}(Z, L^{m-2}).$$

Using the first displayed identification with $t = -3$ proves the claim. $\square$

Proposition 8. *One has*

$$H^2(X, F^* \mathcal{Q}(-3)) \neq 0.$$

Equivalently,

$$H^2(X, \mathcal{E} \otimes \omega_X) \neq 0.$$

Proof. By Lemma 7,

$$H^\bullet(X, \text{Sym}^{p-2} \mathcal{Q}(-2)) = 0.$$

Therefore the short exact sequence (1) induces isomorphisms

$$H^i(X, F^* \mathcal{Q}(-3)) \simeq H^i(X, \text{Sym}^p \mathcal{Q}(-3))$$

for all i .

Taking $m = p$ and $j = 2$ in Lemma 7, we get

$$H^2(X, \text{Sym}^p \mathcal{Q}(-3)) \simeq H^0(Z, L^{p-2}).$$

But $Z = \text{Gr}(3, V) \simeq \mathbb{P}^3$ and $L = \mathcal{O}_Z(1)$, so

$$H^0(Z, L^{p-2}) \neq 0$$

for every prime p , including $p = 2$. Hence

$$H^2(X, F^* \mathcal{Q}(-3)) \neq 0.$$

Finally, by Lemma 3,

$$\mathcal{E} \otimes \omega_X = (F^* \mathcal{Q} \otimes \mathcal{O}_X(1)) \otimes \mathcal{O}_X(-4) = F^* \mathcal{Q}(-3),$$

so the equivalent formulation follows. $\square$

6 Proof of the theorem

We now finish the proof of Theorem 1. By Lemma 4, X is Frobenius split. By Proposition 8,

$$H^2(X, \mathcal{E} \otimes \omega_X) \neq 0.$$

Applying Lemma 2 to $Y = X$ and $\mathcal{A} = \mathcal{E}$ gives

$$H^2(X, F^{n*} \mathcal{E} \otimes \omega_X) \neq 0 \quad \text{for every } n \geq 0.$$

Thus, with

$$\mathcal{G} = \omega_X,$$

we have a fixed coherent sheaf $\mathcal{G}$ such that

$$H^2(X, F^{n*} \mathcal{E} \otimes \mathcal{G}) \neq 0 \quad \text{for every } n \geq 0.$$

Since $\text{rk}(\mathcal{E}) = 2$, this is nonvanishing in the borderline degree $i = \text{rk}(\mathcal{E})$ for all Frobenius iterates. This proves Theorem 1.

Equivalently, because $F^{n*} \mathcal{O}_X(1) = \mathcal{O}_X(p^n)$,

$$F^{n*} \mathcal{E} \otimes \omega_X \simeq F^{(n+1)*} \mathcal{Q} \otimes \mathcal{O}_X(p^n - 4),$$

and the groups

$$H^2(X, F^{(n+1)*} \mathcal{Q}(p^n - 4))$$

are nonzero for every $n \geq 0$.

Remark 9 (Quantitative lower bound). The proof of Lemma 2 shows more than nonvanishing. After restriction of scalars along Frobenius, $H^2(X, \mathcal{E} \otimes \omega_X)$ is a direct summand of $H^2(X, F^{n*} \mathcal{E} \otimes \omega_X)$ for every n . Since the proof of Proposition 8 identifies

$$H^2(X, \mathcal{E} \otimes \omega_X) \simeq H^0(\mathbb{P}^3, \mathcal{O}_{\mathbb{P}^3}(p-2)),$$

one gets

$$\dim_k H^2(X, F^{n*} \mathcal{E} \otimes \omega_X) \geq \binom{p+1}{3}$$

for all $n \geq 0$, up to the harmless Frobenius twist of the k -vector-space structure.

Remark 10 (Why symmetric powers do not force vanishing). There is always an inclusion

$$F^{n*} \mathcal{E} \hookrightarrow \text{Sym}^{p^n} \mathcal{E}$$

locally given by $e \mapsto e^{p^n}$. For an ample vector bundle $\mathcal{E}$, Serre vanishing on $\mathbb{P}_X(\mathcal{E})$ implies that, for a fixed coherent sheaf $\mathcal{H}$,

$$H^j(X, \text{Sym}^m \mathcal{E} \otimes \mathcal{H}) = 0 \quad (j > 0, m \gg 0).$$

The example above shows that this does not imply vanishing for the Frobenius subbundle $F^{n*} \mathcal{E}$. In the present example,

$$H^2(X, \text{Sym}^{p^n} \mathcal{E} \otimes \omega_X) = 0 \quad \text{for } n \gg 0,$$

while

$$H^2(X, F^{n*} \mathcal{E} \otimes \omega_X) \neq 0 \quad \text{for every } n \geq 0.$$

The quotient $\text{Sym}^{p^n} \mathcal{E} / F^{n*} \mathcal{E}$ therefore carries essential cohomological information.